PROJECT CHARTER

1. General Project Information	
Project Name:	**AI-Powered Medical Diagnosis Application**
Impact of project:	Faster diagnostics, reduced workload for healthcare providers, and improved accessibility for underserved regions.

2. Project Team				
	Name	**Department**	**Telephone**	**E-mail**
Project Manager:	Rupesh Bura	Management	+1 314-267-2949	rupesh.bura@slu.edu
Team Members:	Sai Pavan Cheekuri	Data Analytics	+1 646-338-2422	Cheekuri.saipavan16@gmail.com
	Lokesh Mannem	Subject Matter Expert (SME)	+1 732-770-6122	lokesh.mannem@slu.edu
	Sydulu Vemula	Customer	+1 314-917-3301	Sydulu.vemula@slu.edu
	Vydhurya Maheshwaram	Subject Matter Expert (SME)	+1 571-436-6499	Vydhurya.maheshwaram@slu.edu

3. Stakeholders

Healthcare Providers: Doctors, nurses, administrative staff.

Patients: Especially in remote/underserved areas.

Regulatory Bodies: For compliance (HIPAA/GDPR).

4. Project Scope Statement

Project Purpose / Business Justification

Leverage AI (NLP/ML) to improve diagnostic accuracy, reduce delays, and enhance healthcare accessibility.

Objectives (in business terms)

- Achieve 85% diagnostic accuracy.
- Reduce diagnostic time by 40%.
- Deploy in 3 underserved regions within 1 year.

Deliverables

- Symptom input interface.
- AI diagnostic engine with confidence scores.
- Healthcare provider dashboard.

Scope

In-Scope: NLP system, multilingual UI, confidence scoring.

Out-of-Scope: Voice input, full EHR integration, global regulatory approval.

Project Milestones

- Prototype Development (Completed: 02/24/25)
- Testing (03/03/25–03/17/25)
- Beta Deployment (03/24/25–03/31/25)

Major Known Risks (including significant Assumptions)

Risk	Risk Rating (Hi, Med, Lo)
Data privacy	Hi
Model accuracy	Med

Constraints

Budget: 150K–200K. Limited to Synthea synthetic data.

5. Traceability Matrix

Requirement	Information Needed	Output Method	Validation Method	Data Source
85% diagnostic accuracy	Historical diagnoses vs. AI predictions	Text format	Compare AI outputs with expert-validated Synthea data; measure accuracy rate in test environment	Synthea conditions, careplans
40% faster diagnosis time	Time logs (manual vs. AI processing)	Dashboard metric	Benchmark manual diagnosis time vs. AI output time using Synthea encounters timestamps	Synthea encounters
Multilingual UI for accessibility	User language preferences	Dropdown language selector (e.g., English)	Test UI with non-English speakers; verify translations match Synthea patient demographics	Synthea patients
GDPR/HIPAA compliance	Data encryption & access logs	Audit trail in admin dashboard	Third-party security review; validate Synthea data anonymization	Synthea patients
Symptom input (text-only)	Patient-reported symptoms	Text input field with auto-suggestions	Verify symptom-to-disease mapping aligns with Synthea symptom_description and condition_code	Synthea encounters
Healthcare provider dashboard	Aggregated patient/diagnosis data	Interactive charts (e.g., accuracy trends)	Cross-check dashboard metrics with raw Synthea tables (careplans, conditions)	Synthea careplans

6. Communication Strategy

- Weekly status reports to sponsors.
- Bi-weekly team meetings.